

ANSWERS OF MODEL TEST PAPER 7

INTERMEDIATE: GROUP – II

PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

PAPER 6A : FINANCIAL MANAGEMENT

Suggested Answers/ Hints

PART I – Case Scenario based MCQs

1. (b) $K_e = \frac{D_1}{P_0} + g$

$$= \frac{2}{20} + 0.05 = 15\%$$

2. (b) $K_d = \frac{I(1-t) + \frac{(RV-NP)}{n}}{\frac{(RV+NP)}{2}} = \frac{9(1-0.35) + \frac{(100-102.90)}{10}}{\frac{(100+102.90)}{2}} = 5.48\%$

3. (a) $K_p = \frac{PD + \frac{(RV-NP)}{n}}{\frac{(RV+NP)}{2}}$

$$K_p = \frac{11 + \frac{(100-102.82)}{10}}{\frac{(100+102.82)}{2}} = 10.57\%$$

4. (a) Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	(a)	(b)	(c) = (a)×(b)
Debentures (₹ 105 per debenture)	2,88,750	0.1672	0.0548	0.0092
Preference shares (₹ 106 per preference share)	2,38,500	0.1381	0.1057	0.0146
Equity shares (₹ 24)	12,00,000	0.6947	0.1500	0.1042
	17,27,250	1.00		0.1280

WACC (K_o) = 12.80%

$$5. \quad (a) \quad \text{Current Market Price} = \frac{D_1}{K_e - g}$$

$$= \frac{2}{0.10 - 0.05} = ₹ 40 \text{ per share}$$

$$6. \quad (c) \quad DFL = \frac{EBIT}{EBT}$$

$$DFL = 4,00,000/3,00,000 = 1.33$$

$$\text{Interest Coverage Ratio} = \frac{EBIT}{\text{Interest Expense}}$$

$$= 4,00,000/1,00,000 = 4$$

$$\text{Operating Profit Margin} = \frac{\text{Sales}}{EBIT} \times 100$$

$$\text{Operating Profit Margin} = (4,00,000/16,00,000) \times 100 = 25\%$$

$$7. \quad (c) \quad \text{COGS} = \text{Sales} \times (1 - \text{Gross Profit Margin})$$

$$\text{COGS} = 6,00,000 \times (1 - 0.20) = 6,00,000 \times 0.80 = 4,80,000$$

The velocity of stock is 3 months.

$$\text{stock turnovers per year} (12/3) = 4$$

$$\text{Stock Turnover Ratio} = \text{COGS} / \text{Average Stock}$$

$$\text{Average Stock} = 4,80,000/4 = 1,20,000$$

$$\text{Average Stock} = (\text{Opening Stock} + \text{Closing Stock})/2$$

$$\text{Closing Stock} = 1,50,000$$

$$8. \quad (d) \quad 1, 2 \text{ and } 3$$

PART II – Descriptive Questions

1. (a) Balance Sheets of Alpha Limited

Liabilities	₹		Assets	₹	
	31 March 2023	31 March 2024		31 March 2023	31 March 2024
Equity share capital (₹ 10 each fully paid)	20,00,000	20,00,000	Fixed Assets (₹18,90,000– ₹90,000)	18,00,000	15,39,000
Reserve and Surplus (balancing)	1,30,000	1,30,000	Long term investment	—	2,96,600
Profit & Loss A/c (15% of sales)	2,70,000	6,15,600	Current Assets (₹ 10,00,000)		

Current Liabilities			Stock	4,00,000	5,20,000
Bank Overdraft	1,00,000	–	Sundry Debtors	3,00,000	4,95,000
Creditors	3,00,000	4,15,000	Cash at Bank (Balancing)	3,00,000	3,10,000
Total	28,00,000	31,60,600	Total	28,00,000	31,60,600

Calculation for 31st March, 2023

(i) Calculation of Current Liabilities

Suppose that Current Liabilities = x, then current assets will be 2.5 x

Working capital = Current Assets – Current Liabilities

$$6,00,000 = 2.5x - x$$

$$x = 6,00,000 / 1.5 = ₹ 4,00,000 \text{ (C.L.)}$$

Other Current Liabilities = Current Liabilities – Bank Overdraft

$$\text{(Creditors)} = 4,00,000 - 1,00,000 = ₹ 3,00,000$$

$$\text{Current Assets} = 2.5 \times 4,00,000 = ₹ 10,00,000$$

$$(ii) \text{ Liquid Ratio} = \frac{\text{Liquid Assets}}{\text{Current Liabilities}}$$

$$1.5 = \frac{\text{Liquid Assets}}{4,00,000}$$

$$\text{Liquid assets} = ₹ 6,00,000$$

$$\text{Liquid assets} = \text{Current Assets} - \text{Stock}$$

$$6,00,000 = 10,00,000 - \text{Stock}$$

$$\text{So, Stock} = ₹ 4,00,000$$

(iii) Calculation of fixed assets: Fixed assets to proprietary fund is 0.75, working capital is therefore 0.25 of proprietary fund. So,

$$\text{Fixed Assets} = 6,00,000 / 0.25 \times 0.75 = ₹ 18,00,000$$

$$(iv) \text{ Sales} = (14,40,000 / 80) \times 100 = ₹ 18,00,000$$

$$(v) \text{ Debtors} = \frac{2}{12} \times \text{Sales}$$

$$2 / 12 \times 18,00,000 = ₹ 3,00,000$$

$$(vi) \text{ Net profit} = 15\% \text{ of } ₹ 18,00,000 = ₹ 2,70,000$$

Calculation for the year 31st March, 2024

$$(vii) \text{ Sales} = 18,00,000 + (18,00,000 \times 0.2) = 21,60,000$$

(viii) Calculation of fixed assets

	₹		₹
To Opening balance	18,00,000	By Banks (Sale)	90,000
		By Loss on sales of Fixed asset	90,000
		By P & L (Dep.) (5% as in previous year)	81,000
		By Balance b/d	<u>15,39,000</u>
Total	<u>18,00,000</u>		<u>18,00,000</u>

(ix) Net profit for the year 2011, $16\% \times 21,60,000 = ₹ 3,45,600$

Total Profit = $2,70,000 + 3,45,600 = ₹ 6,15,600$

(b) EBIT = ₹ 3,00,000

Less: Interest = ₹ $10,00,000 \times 10\% = ₹ 1,00,000$

Earnings available to equity shareholders = ₹ 2,00,000

Equity capitalization rate = 12.5%

Market value of equity = $\frac{₹ 2,00,000}{12.5\%} = ₹ 16,00,000$

Market value of debt = ₹ 10,00,000

Market value of the firm = ₹ 26,00,000

Overall cost of capital = $\frac{₹ 3,00,000 \times 100}{₹ 26,00,000} = 11.54\%$

(c) (i) **Increase in taxable income if sales increase by 6%.**

Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹ 1,40,000}{₹ 35,000} = 4$

If the sales increases by 6%, EBT will increase by 24%. ($4 \times 6\%$)

(ii) **Increase in EBIT if sales increase by 10%.**

Operating Leverage = $\frac{\text{Contribution}}{\text{Earnings before interest and tax}} = \frac{₹ 1,40,000}{₹ 40,000} = 3.5$

If sales increases by 10%, EBIT will increase by $(3.5 \times 10\%) 35\%$.

(iii) **Increase in taxable income if EBIT increase by 6%.**

Financial Leverage = $\frac{\text{Earnings before interest and tax (EBIT)}}{\text{EBT}} = \frac{₹ 40,000}{₹ 35,000} = 1.14$

If EBIT increases by 6%, EBT will increase by 6.8%. ($1.14 \times 6\%$)

2. (a) Problem mentions that the company has applied to the Private Bank for financing its working capital needs. Ideally, banks would not finance for Depreciation cost being a non-cash cost and it would also not finance the profit for you. So, problem needs to be solved using Cash Cost Basis.

Estimation of working capital required (cash cost basis)

	Particulars	Amount
	A) Current Assets	
	A1) Stock of RM 15,84,960 x 30/360	1,32,080.00
	A2) Stock of WIP (From Cost Statement)	4,77,360.00
	A3) Stock of FG (From Cost Statement)	2,37,500.00
	A4) Debtors 32,74,686 x 45/360	4,09,335.75
	A5) Cash & Cash Equivalents (Given)	1,25,000.00
	Gross Working Capital	13,81,275.75
Less:	B) Current Liabilities	
	B1) Creditors 17,17,040 x 30/360	1,43,086.67
	B2) Lag in Wages Payment 9,20,400 x 15/360	38,350.00
	Excess of Current Assets Over Current Liabilities (A) - (B)	11,99,839.08
Add:	Safety Margin @ 15% Of Net Working Capital	2,11,736.31
	Net Working Capital	14,11,575.39

WN -1: Calculation of Profit

Profit = 25% of total cost i.e 20% of sales price

$$= \{(31,200 - 2,500) \times 150\} \times 20\% = \text{Rs. } 8,61,000$$

WN – 2:

	Completed Units	WIP Units
	31,200	9,360
Raw Mat. Consumed	12,48,000	3,36,960
Direct Wages	7,80,000	1,40,400
Overheads	9,36,000	1,68,480
	29,64,000	6,45,840
Gross Factory Cost	36,09,840	

Add: Op WIP	-
Less: Cl. WIP (At Prime Cost)	4,77,360
Cost of Production	31,32,480
Add: Op FG Stock	-
Less: Cl. FG Stock	2,37,500
Cash Cost of Goods Sold	28,94,980
Add: Selling & Distribution Expenses (Bal. Figure)	3,79,706
Cost Of Sales	32,74,686
Profit*	8,61,000
Sales	41,35,686

*It is assumed that profit is unchanged

WN 3 - Calculation of WIP stock (units) and WIP stock amount

WIP UNITS = 30% of FG produced units i.e 30% of 31,200 units
= 9,360 units

WIP amount (at prime cost)

Raw materials = 9,360 x 40 x 90% = 3,36,960

Direct wages = 9,360 x 25 x 60% = 1,40,400

WN 4 - Calculation of purchases from suppliers

Raw Materials Consumed = OP RM Stock + Purchases - Closing RM Stock

15,84,960 = 0 + Purchases – 1,32,080

Purchases = 17,17,040

WN 5 – Calculation of safety margin

Safety Margin = 15% Of Net Working Capital Needs

Excess Of CA Less CL	85	11,99,839.08
Safety Margin	15	2,11,736.31
Net Working Capital	100	1411575.388

(b) EPS = ROE x BVPS (WN 1)

EPS = 0.15 x 125 = ₹ 18.75

Growth = ROE x Retention Ratio

= 0.15 x 0.65

= 9.75%

D1 = Do (1 + g)

$$= (18.75 \times 35\%)(1 + 0.0975)$$

$$= ₹ 7.20$$

Intrinsic Value of share today - Gordon's Formula

$$P_o = \frac{D_1}{K_e - g}$$

$$= \frac{7.20}{0.20 - 0.0975}$$

$$P_o = ₹ 70.24$$

Intrinsic Value of share today - Walter's Model

$$P_o = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Here D = D₀ assuming it would remain constant through infinity

$$P_o = \frac{6.5625 + \frac{0.15}{0.20}(18.75 - 6.5625)}{0.20}$$

$$P_o = ₹ 78.51$$

WN 1 - Relationship between ROE-EPS-BVPS

$$ROE = \frac{\text{Earnings for Equity Shareholders}}{\text{Equity shareholders funds}}$$

If we divide the numerator and denominator with "No of equity shares"

$$ROE = \frac{\text{Earnings for Equity Shareholders} / \text{No of equity shares}}{\text{Equity shareholders funds} / \text{No of equity shares}}$$

Therefore, ROE = EPS / BVPS

3. Calculation of NPV (Amount in crores)

Year	1	2	3	4	5	6	7	8	9	10
EBT	2.000	2.500	4.000	4.750	6.000	6.400	6.150	5.250	3.800	2.900
Add: Interest	0.195	0.195	0.195	0.252	0.252	0.252	0.252	0.252	0.252	0.252
Add: Allocated Common Cost	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125
Project Profit Before Tax	2.320	2.820	4.320	5.127	6.377	6.777	6.527	5.627	4.177	3.277
Less: Tax	-	-	-	1.154	1.435	1.525	1.469	1.266	0.940	0.737
Profit After Tax	2.320	2.820	4.320	3.973	4.942	5.252	5.058	4.361	3.237	2.539
Add: Depreciation	2.410	2.410	2.410	2.410	2.410	2.410	2.410	2.410	2.410	-
Cash Inflows	4.730	5.230	6.730	6.383	7.352	7.662	7.468	6.771	5.647	2.539

Add: Release Of Working Capital	-	-	-	-	-	-	-	-	-	-	5.000
Add: Net Cash Inflow from sale of asset (Net Of Tax) (WN-3)	-	-	-	-	-	-	-	-	-	-	3.471
Total Cash Inflows	4.730	5.230	6.730	6.383	7.352	7.662	7.468	6.771	5.647	11.010	
DF @ 15%	0.870	0.756	0.658	0.572	0.497	0.432	0.376	0.327	0.284	0.247	
PV Cash Inflow	4.113	3.955	4.425	3.650	3.655	3.312	2.808	2.213	1.605	2.722	

TOTAL PV CI = 32.458 Crores

(-) TOTAL PVCO = 30.000 Crores (Initial Outlay + Working Capital)

NPV = 2.458 Crores

ADVISE - Since NPV is positive, company should go for the project.

- Notes** -
1. Allocated common costs are to be excluded from cash inflows
 2. Dividend distribution are deemed irrelevant for cash flow analysis
 3. Discounting rate = MCLR + premium = 12 + 3 = 15%
 4. Interest exp is to be excluded from the cash inflows as it is already getting covered in the discounting rate above
 5. Professional fees paid for project report and R&D costs being sunk costs are irrelevant for decision making

WN 1 – Calculation of applicable taxes each year

For the first 3 years, tax will be zero and for the next 7 years tax rate applicable would 22.5% (30 x 0.75) as balance tax will be paid in Australia, so it will have no relevance under India perspective calculations.

WN – 2 Calculation of interest expense each year

Since post tax interest rate is given in the question, firstly it needs to be converted to pre-tax rate. However, for the first 3 years of the project, post-tax and pre-tax rate would be same owing to zero taxes

Interest Expense (first 3 years) = 3,00,00,000 X 6.5% = 19,50,000 or 0.195 crores

Interest Expense (next 7 years) = 3,00,00,000 x 8.39% = 25,17,000 or 0.2517 crores

$$\begin{aligned}
 \text{Pre-tax Interest Rate} &= \frac{\text{Post tax Rate}}{1 - \text{India Tax Rate}} \\
 &= 6.5 / (1 - 0.225) \\
 &= \mathbf{8.39\%}
 \end{aligned}$$

WN 3 – CALCULATION OF CAPITAL GAINS INCOME IN YEAR 10

Cost of Asset remaining in the block at the beginning of Year 10

= 3,31,00,000 (2,41,00,000 + 90,00,000)

(+) New Asset purchased during the year = 0

(-) Sale Value of the Asset = 3,50,00,000

Capital Gains Income before tax = 19,00,000

(-) Capital Gains tax = 19,00,000 x 15% = 2,85,000

Net Cash Inflow after tax = 3,50,00,000 - 2,85,000

= 3,47,15,000

B) Current Payback Period = $4 + 1.927 / 7.352$

= 4.262 years

Target Payback Period = 3.5 years

Some key measures to reduce your Payback period are as follows (Only illustrative):

- i. Emphasizing on reduction of operational costs
- ii. Improving marketing thereby resulting into higher sales
- iii. Incorporate product-led growth strategies
- iv. Judicious efforts in bringing down the overall cost of capital thereby reducing the discounting rate and in turn better Payback period.
- v. Leveraging out the presence of the fixed cost

4. (a)

Particulars	Factoring	Forfaiting
A) Meaning	Factoring involves sales of receivables to the financial institution called factor in exchange for immediate cash payment	Forfaiting is a form of export financing where the exporter sells the rights to trade receivables to a forfaiter and receives instant cash
B) Recourse or non-recourse	May be on Recourse or Non-recourse basis	Always non-recourse
C) Amount paid	Firms are generally paid 80% to 90% upfront	100% on the value of exported goods is paid
D) Type of receivables	Receivables may either domestic or international	Receivables are international
E) Cost	Factoring cost in the form of factor commission or fees is to be borne by the seller	Overseas Buyer bears the forfaiting cost, if any

F) Secondary market	Factoring does not involve a secondary market for the receivables, meaning that the transaction is complete once the receivables are sold to the factor.	Forfaiting has a secondary market where the receivables can be traded, enhancing liquidity and providing additional opportunities for investors
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(b) Some of the tasks that demonstrate the importance of good financial management

- Taking care not to over invest in fixed assets
- Balancing cash-outflows with cash-inflows
- Ensuring that there is a sufficient level of working capital
- Setting sales revenue targets that will deliver growth
- Increasing the Gross profit by setting the correct pricing for products or services
- Controlling the level of general and administration expenses by finding more cost-efficient ways of running the day-to-day business operations
- Tax Planning that will minimize the taxes a business has to pay

(c) A drop lock is an arrangement whereby the interest rate on a floating-rate note becomes fixed if it falls to a specified level. Above that level the rate floats based on a benchmark market rate, typically with a semi-annual reset. In other words, drop lock bonds marry the attributes of both floating-rate securities and fixed-rate securities. The drop lock effectively sets a floor on the rate and a guaranteed minimum return to

Or

(c) Advantage to the Company - Stock dividends are suitable in the situation of cash crunch and deficiency faced by the company and suitable when restrictions are imposed by lenders to pay the cash dividend

Advantage to the investor – Improves liquidity in the hands of the investors as bonus shares leads to breaking down of higher priced shares into lower priced shares and hence give a choice to shareholders to sell some of the lower priced shares and get some liquidity